THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Software Testing

LAB TASK 3



Submitted to: Engr.Fahad Nisar

Submitted by: Faizan Shabbir Abbasi

2024-SET-21

Software Testing Lab - Lab Task # 03: Manual Test Execution & Defect Reporting

Report details	
Student name	Faizan Shabbir Abbasi
Roll number	2024-SET-21
Lab instructor	Engr. Fahad Nisar
Semester / session	Fifth, Software Engineering Technology / 2024-28 (SET)
Assigned / due	November 05, 2026 /

1. Application and Test Environment

Item	Details
Application	TodoMatic (todo-react)
GitHub repository	https://github.com/mdn/todo-react
Commit / version	d28218cfb353d474c0cb51b495325db6eb908622 (branch main) / package.json 0.3.0
Same application as Lab 02	Yes. The rules and test cases are unchanged from Lab Task 02.
Operating system	Window 11 pro
Browser and version	Google Chrome, Microsoft Edge
Screen / viewport	Browser DevTools device emulation, Pixel 9, 412 x 924
Node.js / how started	Node.js v24.21.0 ; npm install, npm run dev; opened at http://localhost:3000/todo-react/
Date of execution	November 08, 2026

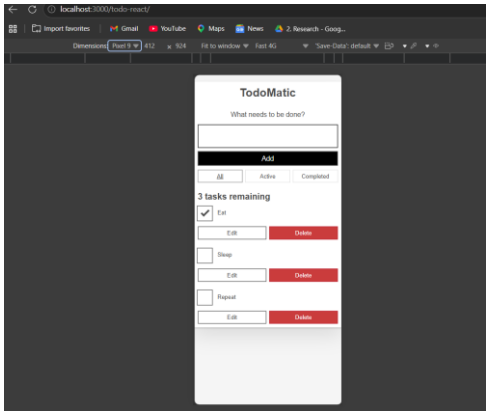
2. Results at a Glance

Test case	Title	Status	Linked defect
TC-01	Application loads with the built-in tasks	**Pass**	-
TC-02	Add a valid task	**Pass**	-
TC-03	Reject an empty new task	**Fail**	DEF-01
TC-04	Reject a spaces-only new task	**Fail**	DEF-01
TC-05	Mark a task completed and view it in the Active filter	**Pass**	-
TC-06	Mark a task active again and check an empty Completed filter	**Pass**	-
TC-07	Rename a task with a valid name	**Pass**	-
TC-08	Reject renaming a task to an empty name	**Fail**	DEF-02
TC-09	Delete a task	**Pass**	-
TC-10	Cancel an edit	**Pass**	-

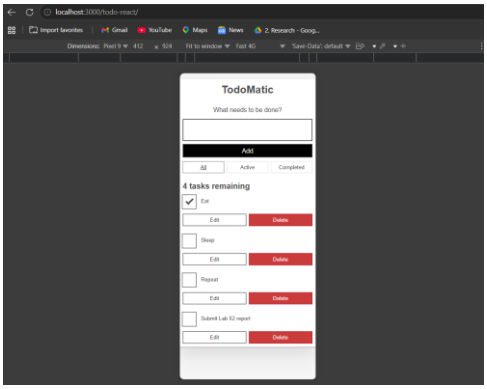
3. Test Case Execution

Each case was started from the three built-in tasks (Eat, Sleep, Repeat) unless a remark says otherwise. Pass = every expected outcome matched; Fail = any mismatch; Blocked = a dependency prevented the check; Not run = not completed. Expected results are copied unchanged from Lab Task 02.

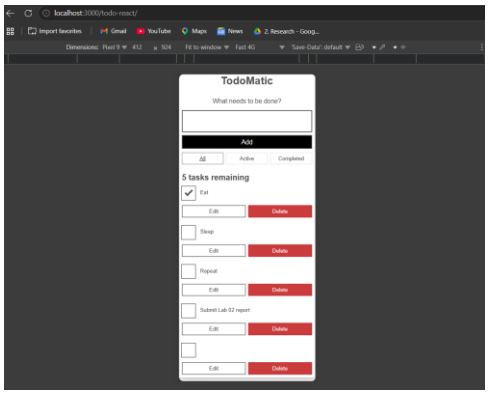
TC-01 - Application loads with the built-in tasks

Field	TC-01 - Application loads with the built-in tasks
Test-case ID / title	TC-01 - Application loads with the built-in tasks
Requirement	R-01: When the application loads, the list contains exactly three tasks in this order: Eat (completed), Sleep (active), Repeat (active). The All filter is selected and the new-task input is empty. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running at http://localhost:3000/todo-react/. Page has just been loaded; nothing has been clicked.
Test data	Built-in tasks: Eat (completed), Sleep, Repeat
Steps	1. Open http://localhost:3000/todo-react/ in a new browser tab. 2. Without clicking anything, inspect the heading, the three filter buttons, the input field and the list.
Expected result	The page title text is "TodoMatic". The All button is pressed (highlighted); Active and Completed are not. The list shows three items in this order: Eat (checkbox ticked), Sleep (unticked), Repeat (unticked), each with Edit and Delete buttons. The count heading reads "3 tasks remaining". The new-task input is empty.
Actual result	The page loaded and showed the heading "TodoMatic". The All filter button was pressed (underlined and boxed); Active and Completed were not. The list showed three items in this order: Eat (checkbox ticked), Sleep (unticked) and Repeat (unticked), each with an Edit and a Delete button. The count heading read "3 tasks remaining" and the new-task input was empty.
Status	**Pass**
Evidence	 LAB03_TC-01.png - initial state after page load
Remarks	All five expected outcomes matched. No defect.

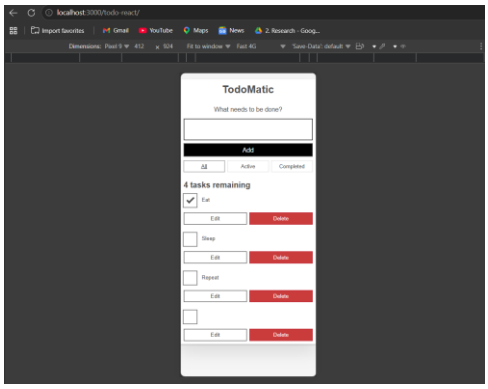
TC-02 - Add a valid task

Field	TC-02 - Add a valid task
Test-case ID / title	TC-02 - Add a valid task
Requirement	R-02: Entering a non-blank name in the new-task input and clicking Add appends one new task to the end of the list, as active (unchecked), with Edit and Delete buttons. The input is cleared. Existing tasks are unchanged. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Submit Lab 02 report
Steps	1. Click the "What needs to be done?" input. 2. Type Submit Lab 02 report. 3. Click Add. 4. Inspect the list, the count heading and the input field.
Expected result	"Submit Lab 02 report" appears exactly once, as the 4th item below Repeat, with an unticked checkbox and Edit and Delete buttons. Eat, Sleep and Repeat are unchanged. The count heading reads "4 tasks remaining". The input field is empty.
Actual result	After typing Submit Lab 02 report in the input and clicking Add, the task appeared once as the 4th item below Repeat, with an unticked checkbox and Edit and Delete buttons. Eat, Sleep and Repeat were unchanged. The heading read "4 tasks remaining" and the input field was empty.
Status	**Pass**
Evidence	 LAB03_TC-02.png - list after adding the task
Remarks	All expected outcomes matched. No defect.

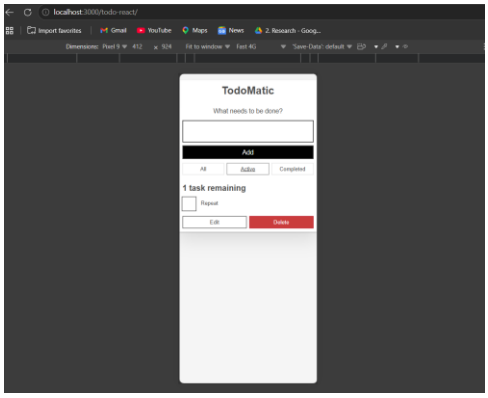
TC-03 - Reject an empty new task

Field	TC-03 - Reject an empty new task
Test-case ID / title	TC-03 - Reject an empty new task
Requirement	R-03: If the new-task input is empty or contains only spaces, clicking Add must not create a task. The task list and the count heading stay as they were.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present. The new-task input is empty.
Test data	(none - input left empty)
Steps	1. Confirm the new-task input contains no text. 2. Click Add. 3. Inspect the list and the count heading.
Expected result	No new item is added; the list still shows only Eat, Sleep and Repeat. The count heading still reads "3 tasks remaining".
Actual result	After clicking Add with the input empty, a new item with a blank name (checkbox, Edit and Delete only) was added at the bottom of the list. The heading increased by one and read "5 tasks remaining" (the list already held 4 tasks, including Submit Lab 02 report from TC-02, before the click).
Status	**Fail**
Evidence	 LAB03_TC-03.png - blank item added after clicking Add with empty input
Remarks	Linked defect: DEF-01 (rule R-03 violated). Note: the page was not reloaded before this run, so the starting list had 4 tasks instead of the 3 in the precondition. The blank task was still accepted; the same defect is shown from a fresh start in TC-04.

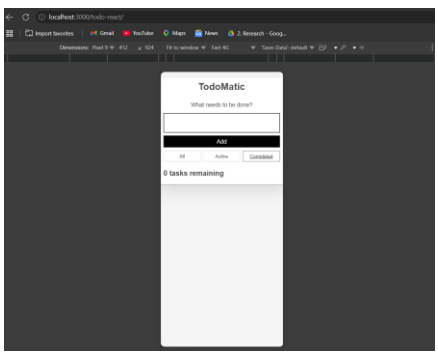
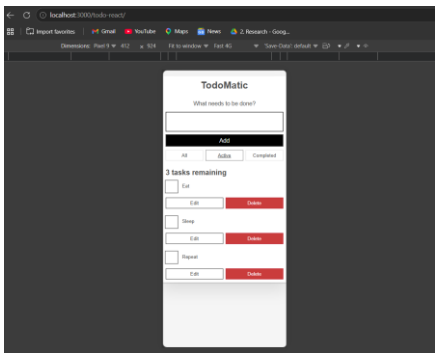
TC-04 - Reject a spaces-only new task

Field	TC-04 - Reject a spaces-only new task
Test-case ID / title	TC-04 - Reject a spaces-only new task
Requirement	R-03: If the new-task input is empty or contains only spaces, clicking Add must not create a task. The task list and the count heading stay as they were. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Three space characters (" ")
Steps	1. Click the new-task input. 2. Press the space bar three times (no other characters). 3. Click Add. 4. Inspect the list and the count heading.
Expected result	No new item (and no item with a blank name) is added; the list shows only Eat, Sleep and Repeat. The count heading still reads "3 tasks remaining".
Actual result	After typing three spaces in the input and clicking Add, a new item with a blank name was added as the 4th item. The heading changed from "3 tasks remaining" to "4 tasks remaining".
Status	**Fail**
Evidence	 LAB03_TC-04.png - blank item added after submitting three spaces
Remarks	Linked defect: DEF-01 (rule R-03 violated).

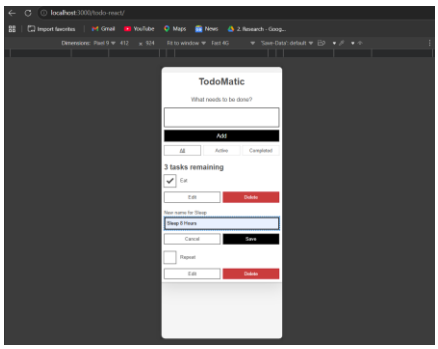
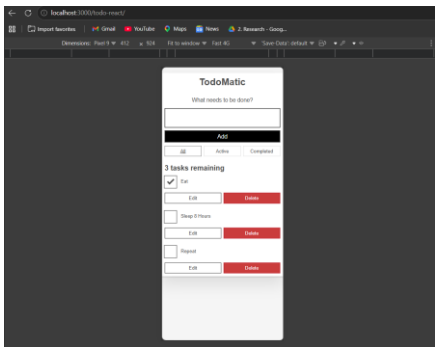
TC-05 - Mark a task completed and view it in the Active filter

Field	TC-05 - Mark a task completed and view it in the Active filter
Test-case ID / title	TC-05 - Mark a task completed and view it in the Active filter
Requirement	R-04: Ticking a task's checkbox marks that task completed; unticking marks it active again. Only that task changes. R-05: The buttons All, Active and Completed are available and exactly one is pressed at a time. All lists every task; Active lists only tasks that are not completed; Completed lists only completed tasks. Switching filters never changes task data. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; Sleep is unticked.
Test data	Task: Sleep Filter: Active
Steps	1. Tick the checkbox next to Sleep. 2. Inspect the list in the All view. 3. Click the Active filter button. 4. Inspect the list, the pressed filter button and the count heading.
Expected result	After step 1, the Sleep checkbox is ticked; Eat is still ticked; Repeat is still unticked. After step 3, only Repeat is listed (Eat and Sleep are not shown). The Active button is pressed. The count heading reads "1 task remaining".
Actual result	After ticking the checkbox next to Sleep and clicking the Active filter, only Repeat was listed; Eat and Sleep were not shown. The Active button was pressed and the heading read "1 task remaining".
Status	**Pass**
Evidence	 LAB03_TC-05.png - Active filter after completing Sleep
Remarks	All expected outcomes matched. No defect.

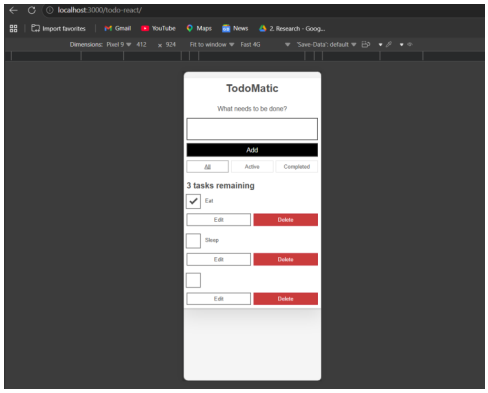
TC-06 - Mark a task active again and check an empty Completed filter

Field	TC-06 - Mark a task active again and check an empty Completed filter
Test-case ID / title	TC-06 - Mark a task active again and check an empty Completed filter
Requirement	R-04: Ticking a task's checkbox marks that task completed; unticking marks it active again. Only that task changes. R-05: The buttons All, Active and Completed are available and exactly one is pressed at a time. All lists every task; Active lists only tasks that are not completed; Completed lists only completed tasks. Switching filters never changes task data. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; Eat is ticked (completed).
Test data	Task: Eat Filters: Completed, then Active
Steps	1. Untick the checkbox next to Eat. 2. Click the Completed filter button and inspect the list and heading. 3. Click the Active filter button and inspect the list and heading.
Expected result	After step 1, the Eat checkbox is unticked. After step 2, the list is empty, the Completed button is pressed and the heading reads "0 tasks remaining". After step 3, Eat, Sleep and Repeat are all listed, the Active button is pressed and the heading reads "3 tasks remaining".
Actual result	After unticking Eat, the Eat checkbox was unticked. After clicking the Completed filter, the list was empty, the Completed button was pressed and the heading read "0 tasks remaining". After clicking the Active filter, Eat, Sleep and Repeat were all listed (none ticked), the Active button was pressed and the heading read "3 tasks remaining".
Status	**Pass**
Evidence	  LAB03_TC-06a.png - Completed filter, empty list LAB03_TC-06b.png - Active filter, all three tasks listed
Remarks	All expected outcomes matched. No defect.

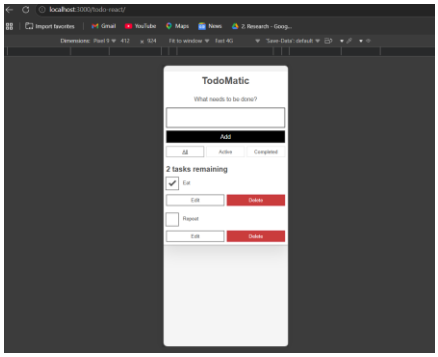
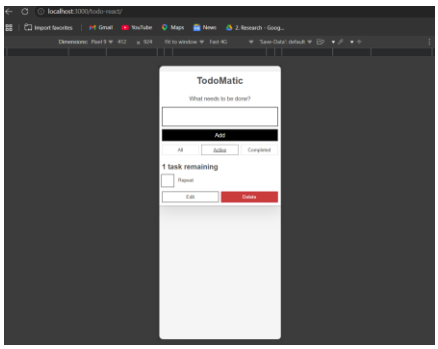
TC-07 - Rename a task with a valid name

Field	TC-07 - Rename a task with a valid name
Test-case ID / title	TC-07 - Rename a task with a valid name
Requirement	R-06: Clicking Edit shows a text field with Cancel and Save. Saving a non-blank name replaces that task's name and keeps its position and completed status. Clicking Cancel discards the typed text and keeps the old name. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to edit: Sleep New name: Sleep 8 Hours
Steps	1. Click Edit next to Sleep. 2. Confirm a text field with Cancel and Save buttons is shown for that task. 3. Click the text field and type Sleep 8 Hours. 4. Click Save. 5. Inspect the list and the count heading.
Expected result	After step 2, the edit form (text field, Cancel, Save) replaces the normal view of the Sleep task only. After step 4, the edit form closes and the second item reads "Sleep 8 Hours" with an unticked checkbox. No item named "Sleep" remains; Eat (ticked) is first and Repeat is third. The count heading reads "3 tasks remaining".
Actual result	After clicking Edit next to Sleep, the edit form (text field, Cancel and Save) replaced the normal view of the Sleep task only. After typing Sleep 8 Hours and clicking Save, the edit form closed and the second item read "Sleep 8 Hours" with an unticked checkbox. No item named "Sleep" remained, Eat (ticked) was first and Repeat was third. The heading read "3 tasks remaining".
Status	**Pass**
Evidence	  LAB03_TC-07a.png - edit form open with the new name typed LAB03_TC-07b.png - list after clicking Save
Remarks	All expected outcomes matched. No defect.

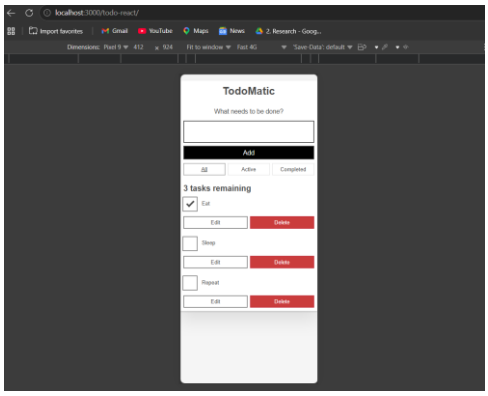
TC-08 - Reject renaming a task to an empty name

Field	TC-08 - Reject renaming a task to an empty name
Test-case ID / title	TC-08 - Reject renaming a task to an empty name
Requirement	R-07: Saving an edit with an empty or spaces-only name must not rename the task; the original name is kept.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to edit: Repeat New name: (none - field left empty)
Steps	1. Click Edit next to Repeat. 2. Leave the text field empty. 3. Click Save. 4. Inspect the list.
Expected result	The task is still named "Repeat"; no item with a blank name is shown. The list shows Eat, Sleep, Repeat in that order and the count heading reads "3 tasks remaining".
Actual result	After clicking Edit next to Repeat, leaving the text field empty and clicking Save, the edit form closed and the task was renamed to a blank name. The list showed Eat, Sleep and an item with no name. The heading still read "3 tasks remaining".
Status	**Fail**
Evidence	 LAB03_TC-08.png - Repeat renamed to a blank name
Remarks	Linked defect: DEF-02 (rule R-07 violated).

TC-09 - Delete a task

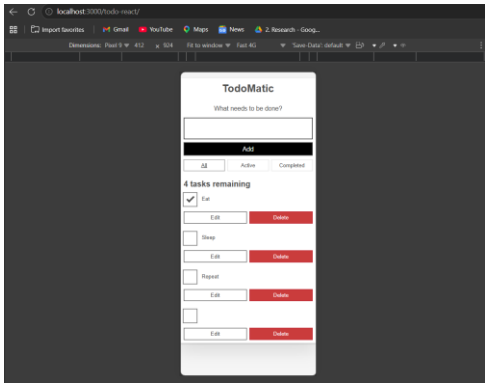
Field	TC-09 - Delete a task
Test-case ID / title	TC-09 - Delete a task
Requirement	R-08: Clicking Delete removes that task permanently from every filter view. Other tasks are unchanged and keep their order. R-05: The buttons All, Active and Completed are available and exactly one is pressed at a time. All lists every task; Active lists only tasks that are not completed; Completed lists only completed tasks. Switching filters never changes task data. R-09: The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to delete: Sleep
Steps	1. Click Delete next to Sleep. 2. Inspect the list and the count heading. 3. Click the Active filter button and inspect the list and heading.
Expected result	After step 1, Sleep is gone; the list shows Eat (ticked) then Repeat (unticked) and the heading reads "2 tasks remaining". After step 3, only Repeat is listed (Sleep does not reappear) and the heading reads "1 task remaining".
Actual result	After clicking Delete next to Sleep, Sleep was removed from the list. The All view showed Eat (ticked) then Repeat (unticked) and the heading read "2 tasks remaining". After clicking Active, only Repeat was listed, Sleep did not reappear, and the heading read "1 task remaining".
Status	**Pass**
Evidence	  LAB03_TC-09a.png - All view after deleting Sleep LAB03_TC-09b.png - Active view after deleting Sleep
Remarks	All expected outcomes matched. No defect.

TC-10 - Cancel an edit

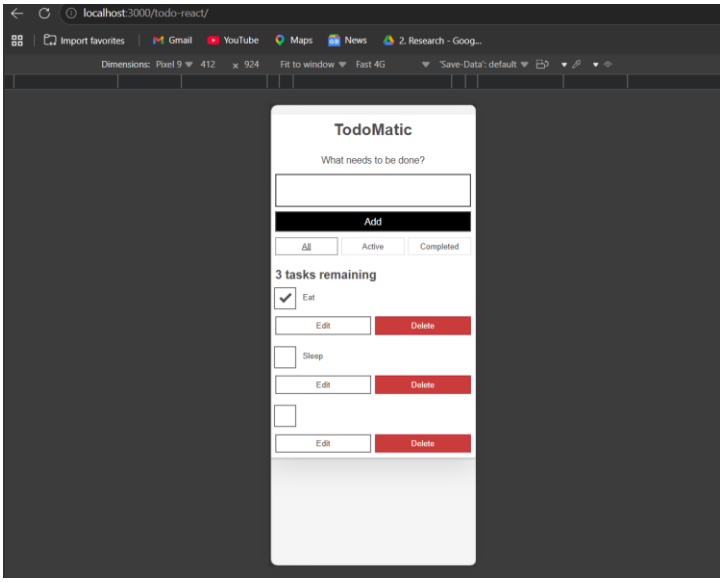
Field	TC-10 - Cancel an edit
Test-case ID / title	TC-10 - Cancel an edit
Requirement	R-06: Clicking Edit shows a text field with Cancel and Save. Saving a non-blank name replaces that task's name and keeps its position and completed status. Clicking Cancel discards the typed text and keeps the old name.
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to edit: Repeat Typed text: Do not save
Steps	1. Click Edit next to Repeat. 2. Type Do not save in the text field. 3. Click Cancel. 4. Inspect the list.
Expected result	The edit form closes and Repeat is shown again in its normal view with an unticked checkbox. The task is still named "Repeat"; the text "Do not save" appears nowhere in the list. The list shows Eat, Sleep, Repeat and the count heading reads "3 tasks remaining".
Actual result	After clicking Edit next to Repeat, typing Do not save and clicking Cancel, the edit form closed. Repeat was shown again in its normal view with an unticked checkbox and kept its name "Repeat". The text "Do not save" did not appear in the list. The list showed Eat, Sleep and Repeat and the heading read "3 tasks remaining".
Status	**Pass**
Evidence	 LAB03_TC-10.png - list after cancelling the edit
Remarks	All expected outcomes matched. No defect.

4. Defect Reports

DEF-01

Field	DEF-01 - The app accepts blank task names when adding a task
Defect ID / title	DEF-01 - The app accepts blank task names when adding a task
Related test case	TC-03 (empty input), TC-04 (spaces-only input)
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application freshly loaded; All view; the three built-in tasks are present.
Steps to reproduce	1. Leave the new-task input empty, or type three spaces. 2. Click Add. 3. Inspect the list and the count heading.
Expected result	Rule R-03: no task is created and the list and the heading stay as they were (3 tasks remaining).
Actual result	A new item with a blank name is added to the list and the heading increases by one (4 tasks remaining from a fresh start).
Evidence	 LAB03_TC-04.png - blank item added (evidence for DEF-01, same as TC-04)

DEF-02

Field	DEF-02 - The app accepts an empty name when renaming a task
Defect ID / title	DEF-02 - The app accepts an empty name when renaming a task
Related test case	TC-08
Version / environment	Commit: d28218cfb353d474c0cb51b495325db6eb908622 Version: 0.3.0 (package.json) Windows, Microsoft Edge, device emulation Pixel 9 (412 x 924)
Preconditions	Application freshly loaded; All view; the three built-in tasks are present.
Steps to reproduce	1. Click Edit next to Repeat. 2. Leave the text field empty. 3. Click Save. 4. Inspect the list.
Expected result	Rule R-07: the task keeps its original name "Repeat".
Actual result	The task is renamed to a blank name and shows as an item with no text.
Evidence	 LAB03_TC-08.png - Repeat renamed to blank (evidence for DEF-02, same as TC-08)

5. Execution Summary

Field	Result
Total test cases	10
Pass	7 (TC-01, TC-02, TC-05, TC-06, TC-07, TC-09, TC-10)
Fail	3 (TC-03, TC-04, TC-08)
Blocked	0
Not run	0
Observed defects	2: DEF-01 (blank task accepted when adding; TC-03, TC-04) and DEF-02 (blank name accepted when renaming; TC-08)
Remaining work	None required. Optional: re-run TC-03 from a freshly loaded page for cleaner evidence (the defect is already confirmed by TC-04).

Check: Pass + Fail + Blocked + Not run = 10.

THANK YOU SO MUCH